

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)							Semester: I		
Course: IT Proficiency							Code: ECVS101		
Teaching Scheme (Hrs/week)				Evaluation Scheme (Marks)					
Lecture	Practical	Tutorial	Credit	CIE	ETE	TW	OR	PR	Total
-	04	-	02	-	-	25	-	-	25
Prerequisites:									
Basic Computer Skills									
Course Objectives:									
1. To develop proficiency in essential office software and tools, including MS Word, MS Excel, MS PowerPoint, and LaTeX, to create, analyze, and present professional documents and data effectively, while understanding ethical internet use and leveraging AI tools.									
Course Outcomes: After completion of this course, students will be able to -									
CO1	Create and format professional documents using MS Word.								
CO2	Organize and analyze data using Excel’s features.								
CO3	Analyze and visualize complex data with pivot tables and charts.								
CO4	Analyze advanced Excel functions, pivot tables, macros, and data protection techniques.								
CO5	Create Professional Documents Using LaTeX.								
CO6	Apply ethical practices in using internet resources and AI tools.								
Course Contents:									
Unit	Description								Duration (Hrs.)
1.	Basics of Computer and MS Word: Awareness of computer Basics MS-Word: Text Basics, Text Formatting and saving file, Working with objects, Header & footers, Working with bullets and numbered lists, Tables, Styles and Content, Merging documents, Sharing and maintaining document, Proofing the document, Printing.								08
2.	MS-Excel: Introduction to Excel, Formatting excel work book, Perform calculations with functions, Sort and Filter data with Excel, Create effective 2D and 3D charts to Present data visually.								10
3.	Advance MS-Excel: Analyze data using pivot tables and pivot charts, Protecting and sharing the work book, Use Macros to automate tasks, Proofing and Printing, More useful functions in excel, Goal seek and scenario features, V-lookup and H-lookup functions, Advanced sort and filter in excel.								10
4.	MS-PowerPoint: Setting up PowerPoint environment, Creating slides and applying themes, Working with bullets and numbering, Working with objects, Hyperlinks and action buttons, Working with movies and sounds, Using SmartArt and Tables, Animation and slide transition, Using slide master, Slide show option, Proofing and Printing.								10
5	Introduction to Latex: Installation of the software LaTeX , Understanding Latex compilation, Basic Syntax, Writing equations, Matrix, Tables. Page Layout – Titles, Abstract Chapters, Sections, References, Equation references, citation. List making environments, Table of contents, Generating new commands, Figure handling, Numbering, List of figures, List of tables, Generating index.								10

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	Packages - Geometry, Hyperref, amsmath, amssymb, algorithms, algorithmic graphic, color, tilez listing. Classes: article, book, report, beamer, slides. IEEtran. Applications - Writing Resume, Writing articles/ research papers, project report.	
6	Internet Ethics & AI tools Working with Internet and-mail, Using the Internet, Internet Ethics and Safety, Social Media. AI Tools: Jasper, GitHub Copilot, Synthesia, Writesonic.	08
TOTAL		56
List of Experiments:		
1. Create a collaborative document project where multiple users contribute to a document using MS Word's track changes and commenting features. 2. To analyze and visualize data effectively using Excel's functions and charts, aiming to create insightful and dynamic data visualizations. 3. Develop a financial modeling project using Excel, incorporating advanced functions like goal seek, scenario analysis, and pivot tables. Build automation using macros for repetitive tasks. 4. Create an interactive multimedia presentation on a complex topic of interest. Incorporate animations, transitions, embedded videos, and interactive elements like hyperlinks and action buttons. 5. Design and implement a digital marketing campaign for a fictitious product or service. Create email newsletters, social media posts, and analyze campaign performance metrics. 6. Prepare research article using Latex.		
Text Books:		
1. Banerjee Snigdha, "MS Word 2000", New Age International. 2. Quentin Docter, Q., et al., "CompTIA IT Fundamentals Study Guide: Exam FC0-U61", Wiley, USA. 3. Lambert, J., Frye, C., et al., "Microsoft Office 2019 Step by Step", Microsoft Press, USA.		
Reference Books:		
1. Walkenbach John, "Excel 2013 Bible", Wiley Publishing House. 2. Wempen Faithe, "Microsoft PowerPoint 2010 Bible", Wiley Publishing House. 3. Miller, M., "Internet Basics Absolute Beginner's Guide", Que Publishing, USA. 4. Miller, M., "Computer Basics Absolute Beginner's Guide", Que Publishing, USA.		
E-Resources:		
1. Microsoft Office Support provides tutorials and guides for MS Office applications. https://support.microsoft.com/en-us/training 2. Digital Skilling by NPTEL - https://elearn.nptel.ac.in/shop/nptel/digital-skilling/?v=c86ee0d9d7ed		